The G/L/C/M classification and shadow tables

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Remark 0.1 (Relationship to companion papers). The three-invariants distinction $p_{\max} \neq k_{\max} \neq r_{\max}$ and the operator-order trichotomy are developed further in the companion paper *The classification trichotomy for chiral algebras* (`classification_trichotomy.tex`).

1 The Four Shadow Classes

The single-line dichotomy (Theorem **??**) partitions the space of chirally Koszul algebras into four shadow classes, determined by two invariants of the shadow metric $Q_L(t) = (2\kappa + 3\alpha t)^2 + 2\Delta\, t^2$: the cubic coefficient $\alpha = S_3$ and the critical discriminant $\Delta = 8\kappa S_4$.

Definition 1.1 (Shadow classes)**.** Let $\mathcal{A}$ be a chirally Koszul algebra with shadow metric Q_L on a one-dimensional primary line L. The *shadow class* of $\mathcal{A}$ along L is:

Class	α	Δ	$r_{\max}$	Q_L
G (Gaussian)	0	0	2	$(2\kappa)^2$
L (Lie/tree)	$\neq 0$	0	3	$(2\kappa + 3\alpha t)^2$
C (Contact)	0	0^*	4	$(2\kappa)^2$ (primary line)
M (Mixed)	$\neq 0$	$\neq 0$	∞	irreducible quadratic

Here $r_{\max}$ is the *shadow depth*: the largest degree r for which the shadow coefficient $S_r \neq 0$. *Class C has $\Delta = 0$ on the primary line (where the single-line dichotomy applies), but the quartic contact invariant $Q^{\mathrm{contact}} \neq 0$ lives on a charged stratum; see Definition 1.4.

The classification is exhaustive: every entry in Table 1 falls into exactly one class, and every class is realized by at least one standard family. The mechanism is algebraic: the shadow generating function $H(t) = t^2\sqrt{Q_L(t)}$ is algebraic of degree 2 (Theorem **??**), and its Taylor coefficients S_r are computed from the convolution recursion $H(t)^2 = t^4 Q_L(t)$, where $H(t) = \sum_{r\geq 2} r\, S_r\, t^r$ and $S_r = a_{r-2}/r$ with a_n the Taylor coefficients of $\sqrt{Q_L}$. When Q_L is a perfect square ($\Delta = 0$), the square root terminates; when Q_L is irreducible ($\Delta \neq 0$), the Taylor expansion is infinite.

1.1 Class G: Gaussian ($r_{\max} = 2$)

Definition 1.2. A chirally Koszul algebra $\mathcal{A}$ is *Gaussian* if $S_r(\mathcal{A}) = 0$ for all $r \geq 3$. Equivalently, $\alpha = 0$ and $\Delta = 0$, so the shadow metric $Q_L(t) = 4\kappa^2$ is constant and the shadow obstruction tower terminates at the scalar curvature.

Examples. Heisenberg $\mathcal{H}_k$ with $\kappa = k$; lattice VOAs V_Λ with $\kappa = \mathrm{rank}(\Lambda)$; free fermion ψ with $\kappa = 1/4$; bc ghosts with $\kappa = -(6\lambda^2 - 6\lambda + 1)$.
Characterization. Class G algebras are freely generated as chiral algebras. Their bar complex is quadratic (the differential has no higher-order terms), and the OPE on each primary line is abelian. The full obstruction is captured by a single scalar:

$$\Theta_{\mathcal{A}} = \kappa(\mathcal{A}) \cdot \eta \otimes \Lambda,$$

where η is the cyclic direction and $\Lambda \in H^*(\bar{\mathcal{M}}_g)$ is the tautological class. All genus-g data is controlled by $F_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g^{\mathrm{FP}}$ (uniform-weight).

1.2 Class L: Lie/tree ($r_{\max} = 3$)

Definition 1.3. A chirally Koszul algebra $\mathcal{A}$ is *Lie/tree* if $\alpha \neq 0$ and $\Delta = 0$, so that $S_3 \neq 0$ but $S_r = 0$ for all $r \geq 4$. The shadow metric is a perfect square: $Q_L(t) = (2\kappa + 3\alpha t)^2$.

Examples. Affine Kac–Moody $V_k(\mathfrak{g})$ for every simple Lie algebra $\mathfrak{g}$ and every non-critical level $k \neq -h^\vee$: type A ($\mathfrak{sl}_N$), types B, C, D (orthogonal and symplectic), exceptional types E_6, E_7, E_8, G_2, F_4. For all of these:

$$\kappa\big(V_k(\mathfrak{g})\big) = \frac{\dim(\mathfrak{g})\,(k + h^\vee)}{2h^\vee}\,.$$

Characterization. The nonzero cubic shadow $S_3 = \alpha$ arises from the Lie bracket (structure constants f_{abc}), while the Jacobi identity forces the quartic shadow $S_4 = 0$ on the primary line. The r-matrix $r(z) = \Omega_{\mathfrak{g}}/((k+h^\vee)z)$ (single simple pole; the bar construction absorbs one power of the OPE pole) encodes all nonlinear structure. The tree-level graph sum terminates at trivalent vertices.

1.3 Class C: Contact ($r_{\max} = 4$)

Definition 1.4. A chirally Koszul algebra $\mathcal{A}$ is *contact* if $\Delta = 0$ on every primary line (so the single-line dichotomy gives $r_{\max}|_L \leq 3$ on each line), but there exists a charged stratum direction with $S_4 \neq 0$ (the quartic contact invariant), and the tower terminates at degree 4: $S_r = 0$ for $r \geq 5$.

Mechanism. Class C escapes the single-line dichotomy (which gives $r_{\max} \in \{2, 3, \infty\}$) via *stratum separation*: the quartic contact invariant Q^{contact} lives on a charged stratum whose self-bracket exits the cyclic deformation complex at degree 5, by rank-one rigidity. On the weight-changing primary line, the tower is abelian ($\alpha = 0$, $S_4 = 0$); the quartic contact is visible only on the full two-dimensional deformation space.
Examples. The $\beta\gamma$ system at conformal weight λ, with

$$\kappa(\beta\gamma_\lambda) = 6\lambda^2 - 6\lambda + 1 = \tfrac{1}{2}\, c(\beta\gamma_\lambda),$$

and its Koszul dual, the bc system at the same weight. At the standard weight $\lambda = 0$ or $\lambda = 1$: $c = 2$ and $\kappa = 1$.

1.4 Class FF: Feigin–Frenkel locus ($\kappa = 0$)

The four classes G, L, C, M stratify the non-critical locus $\kappa(\mathcal{A}) \neq 0$. A fifth class appears at the boundary $\kappa = 0$: the *Feigin–Frenkel locus* where the Sugawara topologisation fails.

Definition 1.5 (Class FF)**.** A chirally Koszul algebra $\mathcal{A}$ belongs to class FF if $\kappa(\mathcal{A}) = 0$. The shadow depth $r_{\max}(\mathcal{A})$ is then unconstrained by the single-line dichotomy and is read from the Feigin–Frenkel coset data $\mathrm{coset}_{FF}(\mathcal{A})$.

Examples. The critical-level affine algebra $V_{-h^\vee}(\mathfrak{g})$ for every simple $\mathfrak{g}$ (so $V_{-2}(\mathfrak{sl}_2)$, $V_{-3}(\mathfrak{sl}_3)$, ...); its commutant lies in the Feigin–Frenkel centre $\mathfrak{z}(\widehat{\mathfrak{g}})$, polynomial in infinitely many Sugawara-like generators. The Koszul dual of class L under the involution $k \mapsto -k - 2h^\vee$ passes through class FF at the fixed point $k = -h^\vee$. Critical $_N$ (at the Feigin–Frenkel image of the affine critical level) is also class FF.
Characterization. At $\kappa = 0$ the conformal vector degenerates: the Sugawara construction $T = (k + h^\vee)^{-1} : J^a J^a :$ has a pole, and topologisation $\mathrm{SC}^{\mathrm{ch,top}} \to E_3^{\mathrm{top}}$ fails. The coset slot of the canonical fingerprint (Definition **??** in the monograph) switches from $(\mathrm{Com}(\mathcal{H}_\kappa, \mathcal{A}), \mathcal{A}/\mathrm{Com})$ to the Feigin–Frenkel coset $\mathrm{coset}_{FF}(\mathcal{A}) \subset \mathfrak{z}(\widehat{\mathfrak{g}})$. Class FF is not a shadow-depth class in the sense of classes G, L, C, M: it is the locus on which the ambient shadow invariants degenerate and the classification is controlled by coset data rather than by (α, Δ).

Remark 1.6 (Why FF is a fifth class, not a refinement)*.* The four classes G, L, C, M are determined by the shadow metric $Q_L(t) = (2\kappa + 3\alpha t)^2 + 2\Delta t^2$, which degenerates at $\kappa = 0$: the leading constant term vanishes, the Riccati algebraicity argument loses its pivot, and the tower of $\Theta_{\mathcal{A}}^{\leq r}$ must be read from a different source. Promoting FF to a fifth companion class is therefore not a refinement of the quaternitomy but a structural completion: the five classes exhaust the full fingerprint stratum (Theorem **??** in the monograph).

1.5 Class M: Mixed ($r_{\max} = \infty$)

Definition 1.7. A chirally Koszul algebra $\mathcal{A}$ is *mixed* if $\alpha \neq 0$ and $\Delta \neq 0$, so the shadow metric Q_L is an irreducible quadratic in t and the shadow obstruction tower does not terminate.

Examples. Virasoro Vir_c with $\kappa = c/2$; $\mathcal{W}$-algebras $\mathcal{W}_N$ with $\kappa = (H_N - 1)\, c$ where $H_N = \sum_{j=1}^{N} 1/j$; Drinfeld–Sokolov reductions of affine Kac–Moody algebras.
Characterization. The shadow growth rate

$$\rho(\mathcal{A}) = \frac{\sqrt{9\alpha^2 + 16\kappa S_4}}{2|\kappa|}$$

controls the asymptotic behaviour of S_r: the coefficients satisfy $S_r \sim C(\mathcal{A})\, \rho(\mathcal{A})^r\, r^{-5/2} \cos(r\theta + \phi)$ for a branch-point argument θ determined by the shadow metric. The critical central charge c^* where $\rho(\mathrm{Vir}_c) = 1$ is the unique positive real root of the cubic

$$5c^3 + 22c^2 - 180c - 872 = 0, \qquad c^* \approx 6.125\,. \tag{1}$$

For $c > c^*$ the tower converges ($\rho < 1$), and for $c < c^*$ it diverges ($\rho > 1$). At the self-dual point $c = 13$: $\rho(\mathrm{Vir}_{13}) \approx 0.467$.

The Virasoro shadow obstruction tower has explicitly computed coefficients through degree 10:

$$S_2 = \tfrac{c}{2}\,, \qquad S_3 = 2\,, \qquad S_4 = \frac{10}{c(5c+22)}\,, \tag{2}$$

$$S_5 = \frac{-48}{c^2(5c+22)}\,, \qquad S_6 = \frac{80(45c+193)}{3c^3(5c+22)^2}\,, \qquad S_7 = \frac{-2880(15c+61)}{7c^4(5c+22)^2}\,. \tag{3}$$

The c-independence of $S_3 = 2$ is now proved algebraically by direct residue extraction; see `virasoro_r_matrix.tex`.

Remark 1.8 (Depth and Koszulness). Shadow depth $r_{\max}$ does *not* characterize chiral Koszulness. All four classes (Gaussian, $r_{\max} = 2$; Lie, $r_{\max} = 3$; contact, $r_{\max} = 4$; mixed, $r_{\max} = \infty$) consist of chirally Koszul algebras. Shadow depth classifies *complexity* within the Koszul world: the operadic complexity conjecture identifies $r_{\max}(\mathcal{A})$ with the A_∞-depth and L_∞-formality level.

Remark 1.9 (Total depth decomposition). The total shadow depth decomposes as $d(\mathcal{A}) = 1 + d_{\mathrm{arith}} + d_{\mathrm{alg}}$ (Theorem **??**), where the algebraic depth $d_{\mathrm{alg}} \in \{0, 1, 2, \infty\}$ and depths ≥ 5 are purely arithmetic (cusp forms contribute beyond the algebraic engine).

2 Shadow Tables

All entries in the tables below are verified against the compute layer. The modular characteristic $\kappa(\mathcal{A})$ is the degree-2 projection of the universal MC element $\Theta_{\mathcal{A}}$, and the genus-g free energy is $F_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g^{\mathrm{FP}}$ for single-generator (uniform-weight) algebras at all genera, and unconditionally at genus 1 for all families. Here $\lambda_g^{\mathrm{FP}} = \frac{2^{2g-1}-1}{2^{2g-1}} \cdot \frac{|B_{2g}|}{(2g)!}$ are the Faber–Pandharipande intersection numbers.

2.1 Master shadow table

Table 1 records, for each standard family, the modular characteristic κ, the shadow class, the shadow depth $r_{\max}$, the cubic coefficient $\alpha = S_3$, the quartic coefficient S_4, and the critical discriminant $\Delta = 8\kappa S_4$.
†On the primary line, $S_4 = 0$ and $\Delta = 0$ for $\beta\gamma$; the quartic contact invariant $Q^{\mathrm{contact}} \neq 0$ lives on a charged stratum (stratum separation, Definition 1.4).
In the table, $H_N = 1 + 1/2 + \cdots + 1/N$ denotes the N-th harmonic number. The Virasoro quartic contact $S_4 = Q^{\mathrm{contact}}_{\mathrm{Vir}} = 10/[c(5c+22)]$ is the unique quartic shadow on the T-line, proved by direct computation from the Virasoro OPE. The T-line shadow data ($S_3 = 2$, $S_4 = 10/[c(5c+22)]$) is *universal*: it coincides for $\mathcal{W}_N$ at all $N \geq 2$, since every $\mathcal{W}_N$ contains the Virasoro subalgebra along the stress-tensor primary line. The additional primary lines (spin-3, spin-4, etc.) carry independent shadow data with separate quartic coefficients.

2.2 Exceptional and non-simply-laced types

Table 2 records the affine Kac–Moody data for exceptional and non-simply-laced types. All are class L.
The kappa formula is universal: $\kappa(V_k(\mathfrak{g})) = \dim(\mathfrak{g})\,(k + h^\vee)/(2h^\vee)$. The central charge sum $c + c' = 2\dim(\mathfrak{g})$ under the Feigin–Frenkel involution $k \mapsto -k - 2h^\vee$, and the complementarity

Table 1: Master shadow table: standard landscape

Algebra $\mathcal{A}$	$\boldsymbol{\kappa(\mathcal{A})}$	**Class**	$\boldsymbol{r_{\max}}$	$\boldsymbol{S_3}$	$\boldsymbol{S_4}$	$\boldsymbol{\Delta}$
Class G (Gaussian)						
Heisenberg $\mathcal{H}_k$	k	G	2	0	0	0
Lattice V_Λ	$\mathrm{rank}(\Lambda)$	G	2	0	0	0
Free fermion ψ	$1/4$	G	2	0	0	0
bc ghosts (weight λ)	$-(6\lambda^2-6\lambda+1)$	G	2	0	0	0
Class L (Lie/tree)						
$\widehat{\mathfrak{sl}}_2$ at level k	$3(k+2)/4$	L	3	$\neq 0$	0	0
$\widehat{\mathfrak{sl}}_N$ at level k	$(N^2-1)(k+N)/(2N)$	L	3	$\neq 0$	0	0
$\widehat{E}_6$ at level k	$13(k+12)/4$	L	3	$\neq 0$	0	0
$\widehat{E}_7$ at level k	$133(k+18)/36$	L	3	$\neq 0$	0	0
$\widehat{E}_8$ at level k	$62(k+30)/15$	L	3	$\neq 0$	0	0
Class C (Contact)						
$\beta\gamma$ (weight λ)	$6\lambda^2-6\lambda+1$	C	4	0	$0^\dagger$	$0^\dagger$
Class M (Mixed)						
Vir_c	$c/2$	M	∞	2	$\dfrac{10}{c(5c+22)}$	$\dfrac{40}{5c+22}$
$\mathcal{W}_3$ at c	$5c/6$	M	∞	$\neq 0$	$\neq 0$	$\neq 0$
$\mathcal{W}_4$ at c	$13c/12$	M	∞	$\neq 0$	$\neq 0$	$\neq 0$
$\mathcal{W}_5$ at c	$77c/60$	M	∞	$\neq 0$	$\neq 0$	$\neq 0$
$\mathcal{W}_N$ at c (general)	$(H_N-1)\,c$	M	∞	$\neq 0$	$\neq 0$	$\neq 0$

$\kappa + \kappa' = 0$ holds for all affine Kac–Moody algebras. For non-simply-laced types, the dual Coxeter number $h^\vee$ differs from the Coxeter number h ($h^\vee = 3$ for B_2 and C_2, while $h = 4$). The shadow class is L regardless: the Jacobi identity kills the quartic on every primary line.

Remark 2.1 (B_2/C_2 isomorphism). The Lie algebras $B_2 = \mathfrak{so}_5$ and $C_2 = \mathfrak{sp}_4$ are isomorphic, sharing dim $= 10$, $h^\vee = 3$, and $h = 4$. Their Cartan matrices are transposes of each other, but all shadow invariants coincide.

2.3 Complementarity table

‡The Koszul dual $\mathcal{H}_k^! = \mathrm{Sym}^{\mathrm{ch}}(V^*)$ is not a Heisenberg algebra: $\kappa(\mathcal{H}_k^!) = -k = \kappa(\mathcal{H}_{-k})$, but the OPE structure of $\mathcal{H}_k^!$ differs from that of $\mathcal{H}_{-k}$. The central charge sum is not defined within a single-parameter family.

Here $K_N = 2(N-1)(2N^2+2N+1)$ is the Koszul conductor for $\mathcal{W}_N = \mathcal{W}(\mathfrak{sl}_N)$ (the sum $c + c'$ under the Feigin–Frenkel involution $k \mapsto -k - 2N$ on the underlying $\mathfrak{sl}_N$). The key structural distinction: for all affine Kac–Moody algebras, $\kappa + \kappa' = 0$ (anti-symmetry under the Feigin–

Table 2: Shadow data for exceptional and non-simply-laced types

$\mathfrak{g}$	$\dim(\mathfrak{g})$	$h^\vee$	$\boldsymbol{\kappa(V_k(\mathfrak{g}))}$	$c(V_k)+c(V_{k'})$	**Class**	$r_{\max}$
$B_2 \cong \mathfrak{so}_5$	10	3	$5(k{+}3)/3$	20	L	3
$C_2 \cong \mathfrak{sp}_4$	10	3	$5(k{+}3)/3$	20	L	3
G_2	14	4	$7(k{+}4)/4$	28	L	3
F_4	52	9	$26(k{+}9)/9$	104	L	3
E_6	78	12	$13(k{+}12)/4$	156	L	3
E_7	133	18	$133(k{+}18)/36$	266	L	3
E_8	248	30	$62(k{+}30)/15$	496	L	3

Frenkel involution); for $\mathcal{W}$-algebras, $\kappa+\kappa' = (H_N-1)\cdot K_N \neq 0$. The Virasoro ($N=2$) gives $\kappa+\kappa' = \frac{1}{2}\cdot 26 = 13$, a nonzero constant independent of c. The Koszul involution $c \mapsto 26-c$ has a fixed point at $c=13$, where $\kappa(\mathrm{Vir}_{13}) = 13/2$ and Vir_{13} is self-dual under uncurved quadratic duality.

2.4 Genus tables

The Faber–Pandharipande numbers and the genus expansion $F_g(\mathcal{A}) = \kappa(\mathcal{A})\cdot\lambda_g^{\mathrm{FP}}$ (uniform-weight).

The formula is $\lambda_g^{\mathrm{FP}} = \frac{2^{2g-1}-1}{2^{2g-1}}\cdot\frac{|B_{2g}|}{(2g)!}$, where B_{2g} is the $2g$-th Bernoulli number. The generating function is $\sum_{g\geq 1}\lambda_g^{\mathrm{FP}}\,\hbar^{2g} = \hat{A}(i\hbar)-1$, where $\hat{A}(x) = (x/2)/\sinh(x/2)$ is the $\hat{A}$-genus (so $\hat{A}(ix) = (x/2)/\sin(x/2)$). The numbers λ_g^{FP} are strictly positive, strictly decreasing, and decay like $(2\pi)^{-2g}$.

The entries for $V_1(\mathfrak{sl}_3)$ use $\kappa = (8\cdot 4)/(2\cdot 3) = 16/3$. The Leech lattice (rank $=24$) gives $F_1 = 1$, reflecting the Jacobi Δ-function origin of the Leech theta series.

2.5 Shadow growth rate table

The shadow growth rate $\rho = \sqrt{(9\alpha^2+16\kappa S_4)/(4\kappa^2)}$ determines the convergence of the shadow obstruction tower. For Virasoro, $\rho^2 = (180c+872)/[c^2(5c+22)]$. The critical central charge $c^* \approx 6.125$ (the positive real root of (1)) separates the convergent regime ($c > c^*$, tower summable) from the divergent regime ($c < c^*$, tower grows exponentially). Class G and class L algebras have $\rho = 0$ (the tower terminates, so there is no exponential growth).

2.6 Virasoro shadow obstruction tower table

Each S_r is a rational function of c with poles at $c=0$ and $c=-22/5$; these are the two values where the quadratic Casimir $c(5c+22)$ vanishes. The sign pattern $S_r > 0$ for even r and $S_r < 0$ for odd $r \geq 5$ (at $c>0$) reflects the oscillatory decay $\cos(r\theta+\phi)$ from the complex branch points of the shadow generating function. All entries have been verified by two independent methods: closed-form derivation and the convolution recursion from $H^2 = t^4 Q_L$.

Table 3: Complementarity data: $\kappa(\mathcal{A}) + \kappa(\mathcal{A}^!)$ and $c(\mathcal{A}) + c(\mathcal{A}^!)$ under Koszul duality

Family $\mathcal{A}$	**Koszul dual $\mathcal{A}^!$**	$\boldsymbol{c + c'}$	$\boldsymbol{\kappa + \kappa'}$	**Self-dual point**
$\mathcal{H}_k$	$\mathrm{Sym}^{\mathrm{ch}}(V^*)$	‡	0	none
V_Λ (lattice)	$V_{\Lambda^!}$	‡	0	Λ self-dual
Free fermion	$\mathrm{Sym}^{\mathrm{ch}}(\gamma)$	0	0	—
$\beta\gamma_\lambda$	bc_λ	0	0	—
$V_k(\mathfrak{sl}_2)$	$V_{-k-4}(\mathfrak{sl}_2)$	6	0	$k = -2$ (critical)
$V_k(\mathfrak{sl}_N)$	$V_{-k-2N}(\mathfrak{sl}_N)$	$2(N^2-1)$	0	$k = -N$ (critical)
$V_k(\mathfrak{g})$ (general)	$V_{-k-2h^\vee}(\mathfrak{g})$	$2\dim(\mathfrak{g})$	0	$k = -h^\vee$ (critical)
Vir_c	Vir_{26-c}	26	13	$c = 13$
$\mathcal{W}_3$ at c	$\mathcal{W}_3$ at $100-c$	100	250/3	$c = 50$
$\mathcal{W}_4$ at c	$\mathcal{W}_4$ at $246-c$	246	533/2	$c = 123$
$\mathcal{W}_5$ at c	$\mathcal{W}_5$ at $488-c$	488	9394/15	$c = 244$
$\mathcal{W}_N$ at c (general)	$\mathcal{W}_N$ at K_N-c	K_N	$(H_N-1)\,K_N$	$c = K_N/2$

Table 4: Faber–Pandharipande intersection numbers λ_g^{FP}

g	λ_g^{FP}	Decimal
1	1/24	$0.041\overline{6}$
2	7/5760	1.215×10^{-3}
3	31/967680	3.204×10^{-5}
4	127/154828800	8.201×10^{-7}
5	73/3503554560	2.084×10^{-8}

3 Structural Theorems

A classification is not a theorem. Left as a table, the four shadow classes would be an observation: a pattern in the data, subject to the next counterexample. What transforms a table into a theorem is the proof that no other classes exist, the identification of the algebraic mechanism that enforces the table, and the discovery that the classes carry structure beyond the taxonomy itself.

This section establishes five results that accomplish this transformation. The depth gap (Proposition 3.1) proves that $d_{\mathrm{alg}} \in \{0, 1, 2, \infty\}$ with no intermediate values: the classification is forced by the arithmetic of a single quadratic polynomial. The SC-formality characterisation (Proposition 3.2) shows that class **G** is distinguished by a sharp operadic property. The betagamma contact mechanism (Proposition 3.4) reveals how class **C** escapes the single-line dichotomy. The DS escalation conjecture (Conjecture 3.6) identifies the functor that moves between classes. The join-semilattice theorem (Proposition 3.8) shows that the classes carry a natural algebraic order.

Table 5: Genus expansion $F_g(\mathcal{A}) = \kappa \cdot \lambda_g^{\mathrm{FP}}$ (uniform-weight) for selected families

Algebra $\mathcal{A}$	$\boldsymbol{\kappa}$	$\boldsymbol{F_1}$	$\boldsymbol{F_2}$	$\boldsymbol{F_3}$
$\mathcal{H}_1$ (Heisenberg, k=1)	1	1/24	7/5760	31/967680
$V_1(\mathfrak{sl}_2)$	9/4	3/32	7/2560	31/430080
$V_1(\mathfrak{sl}_3)$	16/3	2/9	7/1080	31/181440
V_Λ (Leech, rank =24)	24	1	7/240	31/40320
$\beta\gamma$ (standard, λ=1)	1	1/24	7/5760	31/967680
Vir_1	1/2	1/48	7/11520	31/1935360
Vir_{13}	13/2	13/48	91/11520	403/1935360
Vir_{26}	13	13/24	91/5760	403/967680
$\mathcal{W}_3$ at c=6	5	5/24	7/1152	31/193536

Table 6: Shadow growth rate $\rho(\mathcal{A})$ for class M algebras (Virasoro at selected central charges)

c	κ	Q^{contact}	ρ	**Regime**
1/2 (Ising)	1/4	40/49	≈ 12.53	divergent
1	1/2	10/27	≈ 6.24	divergent
4	2	5/84	≈ 1.54	divergent
$c^* \approx 6.125$	≈ 3.06	≈ 0.031	1	critical
13 (self-dual)	13/2	10/1131	≈ 0.467	convergent
25	25/2	2/735	≈ 0.242	convergent
26 (bosonic string)	13	5/1976	≈ 0.232	convergent

3.1 The depth gap theorem

The shadow tower of a chirally Koszul algebra can terminate or diverge. One might expect every finite depth to be realised: that the tower could stop at degree 5, or 7, or 100, by some suitable choice of OPE data. This does not happen. The tower terminates at degree 3 (class **G**), at degree 4 (class **L**), at degree 5 (class **C**, via stratum separation), or not at all (class **M**). No other depth is accessible.

The reason is that the shadow generating function $H(t) = t^2\sqrt{Q_L(t)}$ is controlled by a quadratic polynomial $Q_L(t)$, and a quadratic polynomial is either a perfect square or it is not. When Q_L is a perfect square, the square root is linear and the tower terminates; when Q_L is not a perfect square, the binomial series for $(1+u)^{1/2}$ produces infinitely many nonzero coefficients. There is no third possibility. This is the Riccati dichotomy: the depth of the shadow tower is not a free parameter but is read off from the discriminant of a single quadratic.

Two distinct notions of depth appear in this analysis and must not be conflated. The algebraic depth $d_{\mathrm{alg}}(\mathcal{A})$ counts independent shadow coefficients: $d_{\mathrm{alg}} = 0$ if $S_r = 0$ for all $r \geq 3$, $d_{\mathrm{alg}} = n$ if $S_r = 0$ for all $r \geq n+3$, and $d_{\mathrm{alg}} = \infty$ if infinitely many S_r are nonzero. The generating depth $d_{\mathrm{gen}}(\mathcal{A})$ is the degree at which higher operations are recursively determined by

Table 7: Virasoro shadow coefficients S_r for $r = 2, \ldots, 10$

r	$S_r(\mathrm{Vir}_c)$
2	$c/2$
3	2
4	$10/[c(5c+22)]$
5	$-48/[c^2(5c+22)]$
6	$80(45c+193)/[3c^3(5c+22)^2]$
7	$-2880(15c+61)/[7c^4(5c+22)^2]$
8	$80(2025c^2+16470c+33314)/[c^5(5c+22)^3]$
9	$-1280(2025c^2+15570c+29554)/[3c^6(5c+22)^3]$
10	$256(91125c^3+1050975c^2+3989790c+4969967)/[c^7(5c+22)^4]$

lower ones. For the Virasoro algebra, $d_{\mathrm{gen}} = 3$ (the cubic $S_3 = 2$ and quartic $S_4 = 10/[c(5c{+}22)]$ determine all S_r via the convolution recursion) but $d_{\mathrm{alg}} = \infty$ (every S_r is nonzero). The recursion is finitely generated; the output is not finitely supported.

Proposition 3.1 (Depth gap)**.** *Let $\mathcal{A}$ be a chirally Koszul algebra in the standard landscape with $\kappa(\mathcal{A}) \neq 0$. Then*

$$d_{\mathrm{alg}}(\mathcal{A}) \;\in\; \{0,\, 1,\, 2,\, \infty\}.$$

No finite value $d_{\mathrm{alg}} \geq 3$ is realized. The four values correspond bijectively to the shadow classes **G**, **L**, **C**, **M**.

Proof. The argument splits into a rank-one analysis on each primary line and a multi-sector extension.

Rank one. On a primary line L with $\kappa \neq 0$, the shadow generating function $H(t) = t^2\sqrt{Q_L(t)}$ is determined by (κ, α, S_4) (Theorem **??**), where the shadow metric $Q_L(t) = (2\kappa + 3\alpha\, t)^2 + 2\Delta\, t^2$ is the Gaussian decomposition and $\Delta = 8\kappa S_4$ is the critical discriminant.

If $\Delta = 0$ (equivalently $S_4 = 0$), then $Q_L(t) = (2\kappa + 3\alpha\, t)^2$ is a perfect square, and $H(t) = t^2(2\kappa + 3\alpha\, t)$ is a polynomial of degree 3. The shadow coefficients $S_r = 0$ for all $r \geq 4$, giving $d_{\mathrm{alg}} \in \{0, 1\}$ according to whether $\alpha = 0$ (class **G**) or $\alpha \neq 0$ (class **L**).

If $\Delta \neq 0$ (equivalently $S_4 \neq 0$), then Q_L is not a perfect square in $k(c)[t]$: the interaction correction $2\Delta\, t^2$ prevents $\sqrt{Q_L}$ from being polynomial. The Taylor expansion of $\sqrt{Q_L(t)}$ about $t = 0$ has infinitely many nonzero coefficients (the binomial series for $(1+u)^{1/2}$ with $u = (6\alpha\, t + (9\alpha^2 + 2\Delta)\, t^2)/(4\kappa^2)$ does not terminate when the coefficient of t^2 in u is nonzero). Therefore $S_r \neq 0$ for infinitely many r, and $d_{\mathrm{alg}} = \infty$ (class **M**).

The gap at $d_{\mathrm{alg}} = 3$ is now forced. Suppose $S_4 \neq 0$. The convolution recursion at degree 5 gives $S_5 = -(6/5)(1/\kappa)\, \alpha\, S_4$. If $\alpha \neq 0$: this is nonzero, so $S_5 \neq 0$. If $\alpha = 0$: then $S_5 = 0$, but the degree-6 recursion gives $S_6 \propto S_4^2/\kappa \neq 0$. In either case, $S_4 \neq 0$ propagates to infinitely many nonzero S_r: the closed form $H(t) = t^2\sqrt{Q_L(t)}$ makes this manifest. Termination at any finite degree ≥ 5 is impossible.

Class **C** ($d_{\mathrm{alg}} = 2$) occupies the boundary where $\Delta = 0$ on the primary line but $S_4 \neq 0$ on a charged stratum: stratum separation (Remark **??**) terminates the tower at degree 4 by a mechanism external to the rank-one Riccati analysis. See Proposition 3.4 below.

Multi-sector extension. For a multi-generator algebra with primary lines $L_1, \ldots, L_n$, the cubic pump propagates: if $\alpha \neq 0$ on any mixed sector, the adjoint ad_{S_3} generates an infinite chain on every reachable sector, forcing $d_{\mathrm{alg}} = \infty$. If all mixed cubics vanish, the sectors decouple and $d_{\mathrm{alg}} \leq \max_i d_{\mathrm{alg}}(L_i) \leq 2$. No intermediate value $d_{\mathrm{alg}} = 3$ is accessible. □

3.2 SC-formality characterises class G

Every algebra in the standard landscape is chirally Koszul: the bar spectral sequence collapses at E_2, the bar cohomology concentrates in degree 1, and the transferred A_∞ operations on $H^*(\bar{B}^{\mathrm{ch}}(\mathcal{A}))$ satisfy $m_k = 0$ for $k \geq 3$. Koszulness does not distinguish the four shadow classes.

A stronger condition does. Swiss-cheese formality asks that the higher SC-operations on $\mathcal{A}$ *itself* vanish: $m_k^{\mathrm{SC}} = 0$ for all $k \geq 3$. This is a condition on the algebra, not on its bar cohomology; the common notation m_k for both is a source of systematic confusion. The question is: which algebras satisfy this stronger condition?

The answer is a unique-survivor argument. Four shadow classes enter; one constraint eliminates three.

Proposition 3.2 (SC-formality iff class **G**)**.** *Let $\mathcal{A}$ be a chiral algebra in the standard landscape. Then $\mathcal{A}$ is Swiss-cheese formal if and only if $\mathcal{A}$ belongs to class* **G***.*

Proof. The forward direction is immediate. If $\mathcal{A}$ is class **G** (Heisenberg, lattice, free fermion, *bc* ghosts), the current algebra bracket is central: $[a_m, a_n] = km\,\delta_{m+n,0}$ for Heisenberg, and the analogous statement for every other class **G** family. All nested brackets vanish, so $m_k^{\mathrm{SC}} = 0$ for $k \geq 3$.

The converse is a three-step forced march. Suppose $\mathcal{A}$ is SC-formal. The cubic shadow S_3 is the normalized projection of the trilinear form

$$C(x, y, z) \;=\; \kappa\big(x,\, [y, z]\big),$$

where κ is the invariant bilinear form and $[-,-]$ the current algebra bracket. The chain of implications is:

$$\text{SC-formal} \implies m_3^{\mathrm{SC}} = 0 \implies S_3 = 0 \implies C = 0.$$

The first arrow is the definition. The second holds because S_3 is the degree-3 shadow projection of m_3^{SC}. The third requires the key input: κ is nondegenerate on the standard landscape. With κ nondegenerate, $C = 0$ forces $[y, z] = 0$ for all y, z. The current algebra is abelian; an abelian current algebra at level k is Heisenberg; Heisenberg is class **G**.

Classes **L**, **C**, and **M** all have $S_3 \neq 0$ or carry structure (charged strata, nonlinear composites) incompatible with $m_3^{\mathrm{SC}} = 0$. Each is eliminated. Only class **G** survives. □

Remark 3.3 (The two meanings of formality). The distinction between A_∞-formality of bar cohomology and SC-formality of the algebra itself is the central subtlety of the Koszulness programme. Every chirally Koszul algebra is A_∞-formal on its bar cohomology; this is a consequence of the E_2-collapse and holds in all four shadow classes. SC-formality is a strictly stronger condition, operative on the algebra rather than on its cohomological shadow. Proposition 3.2 identifies the precise locus where the two notions coincide: class **G**.

3.3 The betagamma contact mechanism

The single-line Riccati dichotomy produces three outcomes: $d_{\mathrm{alg}} \in \{0, 1, \infty\}$. Class **C** ($d_{\mathrm{alg}} = 2$) is absent from this list. Its existence requires a mechanism that the rank-one analysis cannot see: shadow data that lives on a charged stratum rather than on the primary line, and a termination mechanism that operates by dimensional obstruction rather than by algebraic identity.

The $\beta\gamma$ system is the algebra that exhibits this mechanism. It is the unique standard family where shadow depth exceeds collision depth: the OPE pole order is $p_{\max} = 1$ (a single simple pole $\beta(z)\gamma(w) \sim 1/(z-w)$), the collision depth is $k_{\max} = 0$ (no collision singularity), yet the shadow depth is $r_{\max} = 4$. Three invariants that agree for every other standard family here separate: $p_{\max} \neq k_{\max} \neq r_{\max}$. The mechanism is stratum separation: the quartic shadow lives on a charged stratum invisible to the single-line dichotomy, and the tower terminates there by rank-one rigidity.

Proposition 3.4 (Contact class via stratum separation)**.** *Let $\beta\gamma_\lambda$ denote the $\beta\gamma$ system at conformal weight λ, with central charge $c(\beta\gamma_\lambda) = 2(6\lambda^2 - 6\lambda + 1)$ and $\kappa(\beta\gamma_\lambda) = 6\lambda^2 - 6\lambda + 1$. Then*:

(i) *On the weight-changing primary line, the OPE is abelian ($\alpha = 0$) and $\Delta = 0$: the single-line dichotomy gives $r_{\max}|_L = 2$ (class **G** on this line).*

(ii) *The quartic contact invariant $\mathfrak{Q}_{\beta\gamma} = \mathrm{cyc}(m_3)$ is nonzero and lives in the charged sector $\mathrm{Def}^{(q)}_{\mathrm{cyc}}$ with $q \neq 0$.*

(iii) *The self-bracket $\mathrm{Def}^{(q)}_{\mathrm{cyc}} \otimes \mathrm{Def}^{(q)}_{\mathrm{cyc}} \to \mathrm{Def}^{(2q)}_{\mathrm{cyc}}$ has $\dim \mathrm{Def}^{(2q)}_{\mathrm{cyc}} = 0$ for rank-one systems: the quartic pump does not activate, and the tower terminates at $r_{\max} = 4$.*

(iv) *The shadow depth is $r_{\max}(\beta\gamma_\lambda) = 4$ and $d_{\mathrm{alg}}(\beta\gamma_\lambda) = 2$ for all λ.*

Proof. The four items form a logical chain: (i) shows that the primary line sees nothing beyond class **G**; (ii) identifies where the quartic shadow lives; (iii) explains why the tower stops there; (iv) assembles the classification.

Item (i): invisibility on the primary line. The $\beta\gamma$ OPE $\beta(z)\gamma(w) \sim 1/(z-w)$ is a simple pole. The self-OPEs $\beta(z)\beta(w)$ and $\gamma(z)\gamma(w)$ are regular. There is no current algebra bracket: the OPE on the weight-changing line is abelian, $\alpha = 0$, $S_4 = 0$, $\Delta = 0$. The Riccati dichotomy applied to this line gives $d_{\mathrm{alg}} = 0$. If the primary line were the whole story, $\beta\gamma$ would be class **G**.

Item (ii): birth of the quartic on the charged stratum. The primary line is not the whole story. The $\beta\gamma$ system has two generators of different conformal weight: the cyclic deformation complex decomposes by charge, and the charged sector $\mathrm{Def}^{(q)}_{\mathrm{cyc}}$ with $q \neq 0$ is invisible to the restriction $q = 0$ that defines the primary line. The quartic contact invariant $\mathfrak{Q}_{\beta\gamma} = \mathrm{cyc}(m_3)$ is the trilinear Massey product on this charged stratum (Theorem **??**). At conformal weight $\lambda = 1/2$ (the symplectic boson), a $\mathbb{Z}/2$ symmetry forces $\mathfrak{Q} = 0$; at all other λ, this symmetry is broken and $\mathfrak{Q} \neq 0$. The quartic shadow is born.

Item (iii): death of the quintic by rank-one rigidity. The quartic shadow, once born, must either propagate (producing $d_{\mathrm{alg}} = \infty$, class **M**) or die (giving $d_{\mathrm{alg}} = 2$, class **C**). The propagation mechanism is the quartic pump: the self-bracket $\mathrm{Def}^{(q)}_{\mathrm{cyc}} \otimes \mathrm{Def}^{(q)}_{\mathrm{cyc}} \to \mathrm{Def}^{(2q)}_{\mathrm{cyc}}$. For rank-one systems (one generator in each charge sector), the target space has dimension zero: there is no field of charge $2q$ in a rank-one system. The quintic obstruction S_5 vanishes identically on the charged stratum; the pump has no room to operate. The tower terminates.

Item (iv). Items (i)–(iii) give $S_r = 0$ for $r \geq 5$ on every stratum and $S_4 \neq 0$ on the charged stratum: $r_{\max} = 4$, $d_{\mathrm{alg}} = 2$. The result holds for all λ (the rank-one rigidity is independent of conformal weight), confirming class **C**. □

Remark 3.5 (Why contact is rare). The mechanism that produces class **C** requires three simultaneous conditions: an abelian primary line ($\alpha = 0$), a nonzero quartic on a charged stratum ($\mathfrak{Q} \neq 0$), and rank-one rigidity to terminate the tower ($\dim \mathrm{Def}^{(2q)} = 0$). The third condition fails for rank ≥ 2: the target of the quartic pump acquires positive dimension, and the tower generically diverges. Among standard families, only the $\beta\gamma$ system and its Koszul dual bc satisfy all three conditions.

3.4 Shadow depth under Drinfeld–Sokolov reduction

The four shadow classes are static invariants of individual algebras. A natural question is how these invariants transform under the functors that connect families. Quantum Drinfeld–Sokolov reduction is the principal such functor: it sends an affine Kac–Moody algebra $V_k(\mathfrak{g})$ to a $\mathcal{W}$-algebra $\mathcal{W}^k(\mathfrak{g}, f)$, parameterised by a choice of nilpotent $f \in \mathfrak{g}$. At $f = 0$ the reduction is trivial and the output is the input. At $f \neq 0$ the output is a genuinely different algebra: new composite fields appear (the Sugawara stress tensor T, higher-spin currents W_s), and these composites carry OPE data absent from the affine algebra.

The shadow-theoretic consequence is that DS reduction *escalates* shadow class. An affine Kac–Moody algebra is class **L** ($r_{\max} = 3$): the Jacobi identity kills the quartic shadow on every primary line. Every non-trivial DS reduction creates a Virasoro subalgebra whose quartic shadow $S_4 = 10/[c(5c{+}22)] \neq 0$ on the stress-tensor line, forcing $\Delta \neq 0$ and hence $r_{\max} = \infty$ by the depth gap. The jump is maximal: from $r_{\max} = 3$ (finite tower, class **L**) to $r_{\max} = \infty$ (infinite tower, class **M**), with no intermediate stop.

Conjecture 3.6 (Shadow depth escalation under DS reduction)**.** *Let $\mathfrak{g}$ be a simple Lie algebra, $f \in \mathfrak{g}$ a non-zero nilpotent, and $k \neq -h^\vee$ a non-critical level. Then*

$$r_{\max}\big(\mathcal{W}^k(\mathfrak{g}, f)\big) \;\geq\; r_{\max}\big(V_k(\mathfrak{g})\big) \;=\; 3.$$

*The inequality is strict for every non-zero nilpotent: $r_{\max}(\mathcal{W}^k(\mathfrak{g}, f)) = \infty$ (class **M**) whenever $f \neq 0$.*

The conjecture asserts that DS reduction is a one-way door in shadow complexity: once the nilpotent f is turned on, the tower cannot be made to terminate again. The depth gap (Proposition 3.1) reduces the conjecture to a single computation: it suffices to show $S_4 \neq 0$ on the stress-tensor primary line of $\mathcal{W}^k(\mathfrak{g}, f)$. The infinite tower then follows automatically.

Remark 3.7 (Evidence). Three independent lines of evidence support the conjecture.

(a) Principal reductions. For $f = f_{\mathrm{prin}}$ and every simple $\mathfrak{g}$, the conjecture is proved: the principal $\mathcal{W}$-algebra $\mathcal{W}_N = \mathcal{W}^k(\mathfrak{sl}_N, f_{\mathrm{prin}})$ contains the Virasoro algebra as a subalgebra, and the Virasoro quartic $S_4 \neq 0$ on the T-line is a universal computation independent of $\mathfrak{g}$.

(b) Non-principal reductions in type A. Systematic computation verifies the conjecture for all nilpotent orbits of $\mathfrak{sl}_N$ ($N = 3, 4, 5$) at multiple non-critical levels k. Every non-trivial orbit (minimal, subregular, hook-type, and all intermediate) produces class **M**. The minimal nilpotent orbit of $\mathfrak{sl}_3$ (the Bershadsky–Polyakov algebra $\mathcal{W}^k(\mathfrak{sl}_3, f_{\min})$) is the simplest non-principal example: it has a single strong generator beyond the stress tensor, and $S_4 \neq 0$ is verified by direct OPE computation.

(c) Structural mechanism. Every non-trivial nilpotent f produces, via the BRST cohomology of the associated quantum DS complex, at least one composite strong generator whose self-OPE has pole order ≥ 4. The pole order of the composite is bounded below by the Sugawara construction: the stress tensor T always appears as a strong generator, and $T(z)T(w)$ has a quartic pole. This quartic pole creates $S_4 \neq 0$ on the stress-tensor primary line, and the depth gap converts this single nonzero quartic into an infinite tower.

3.5 The join-semilattice of shadow classes

The four shadow classes are invariants of individual algebras. The question of how they behave under algebraic operations requires a separate argument. Direct sum is the simplest such operation: given two chiral algebras $\mathcal{A}_1$ and $\mathcal{A}_2$, their direct sum $\mathcal{A}_1 \oplus \mathcal{A}_2$ has fields from both factors, with OPE between distinct factors regular (no singular terms). What is the shadow class of the direct sum?

The answer is the maximum, and the proof is that the shadow tower decomposes sector by sector when the cross-sector OPE is regular. This gives the four shadow classes the structure of a join-semilattice: a partially ordered set in which every pair of elements has a least upper bound, realised concretely by direct sum.

Proposition 3.8 (Shadow classes form a join-semilattice)**.** *The four shadow classes* $\{\mathbf{G}, \mathbf{L}, \mathbf{C}, \mathbf{M}\}$ *form a join-semilattice under the total order* $\mathbf{G} < \mathbf{L} < \mathbf{C} < \mathbf{M}$*, where the join is realised by direct sum of chiral algebras:*

$$\mathrm{class}(\mathcal{A}_1 \oplus \mathcal{A}_2) \;=\; \max\bigl(\mathrm{class}(\mathcal{A}_1),\, \mathrm{class}(\mathcal{A}_2)\bigr). \tag{4}$$

Equivalently, the shadow depth satisfies $r_{\max}(\mathcal{A}_1 \oplus \mathcal{A}_2) = \max\bigl(r_{\max}(\mathcal{A}_1),\, r_{\max}(\mathcal{A}_2)\bigr)$.

Proof. The OPE of $\mathcal{A}_1 \oplus \mathcal{A}_2$ decomposes into three sectors: pure $\mathcal{A}_1$, pure $\mathcal{A}_2$, and mixed. The mixed sector has regular OPE by the definition of direct sum: fields from distinct factors are mutually local with no singular terms. Regular OPE contributes no shadow coefficients at any degree. The shadow tower of $\mathcal{A}_1 \oplus \mathcal{A}_2$ is therefore the direct sum of the towers of $\mathcal{A}_1$ and $\mathcal{A}_2$, and the shadow depth is the maximum of the two.

The decisive case is $\mathbf{L} \oplus \mathbf{L}$: does combining two affine Kac–Moody factors produce a quartic shadow? The answer is no. The quartic obstruction S_4 is intrinsic to each factor's OPE. On the primary line of $V_k(\mathfrak{g}_1)$, the Jacobi identity kills S_4; on the primary line of $V_{k'}(\mathfrak{g}_2)$, the same. The mixed channel $\mathfrak{g}_1 \otimes \mathfrak{g}_2$ contributes no singular OPE terms, so no quartic shadow is born in the cross-sector. Therefore $\mathbf{L} \oplus \mathbf{L} = \mathbf{L}$ (not $\mathbf{C}$, not $\mathbf{M}$): the Jacobi identity of each factor survives the direct sum. □

Remark 3.9 (Monotonicity and its failure). The join-semilattice property is specific to direct sums: the functor $\oplus$ preserves the regularity of cross-sector OPE. Constructions that create new singular terms between sectors violate monotonicity and can raise the shadow class strictly. Two examples illustrate the failure.

Drinfeld–Sokolov reduction: $V_k(\mathfrak{g})$ is class $\mathbf{L}$, but $\mathcal{W}^k(\mathfrak{g}, f)$ is class $\mathbf{M}$ for every non-zero nilpotent f (Conjecture 3.6, proved for principal reductions). The BRST construction creates composite fields with new OPE singularities; these singularities produce the quartic shadow that the Jacobi identity had killed.

Coset construction: $\mathcal{A}/\mathcal{A}'$ can land in a higher class than either factor. The Goddard–Kent–Olive coset of $V_k(\mathfrak{sl}_2)$ by its Cartan subalgebra produces the parafermion algebra, whose shadow depth depends on the coset data rather than on the shadow classes of the two inputs.

Shadow depth is monotone under direct sum, anti-monotone under no operation, and unconstrained under quotient or reduction. The join-semilattice records the stable part of this structure; the unstable part is the subject of Conjecture 3.6.